Finite Automata: Deterministic (DFA) and Nondeterministic (NFA

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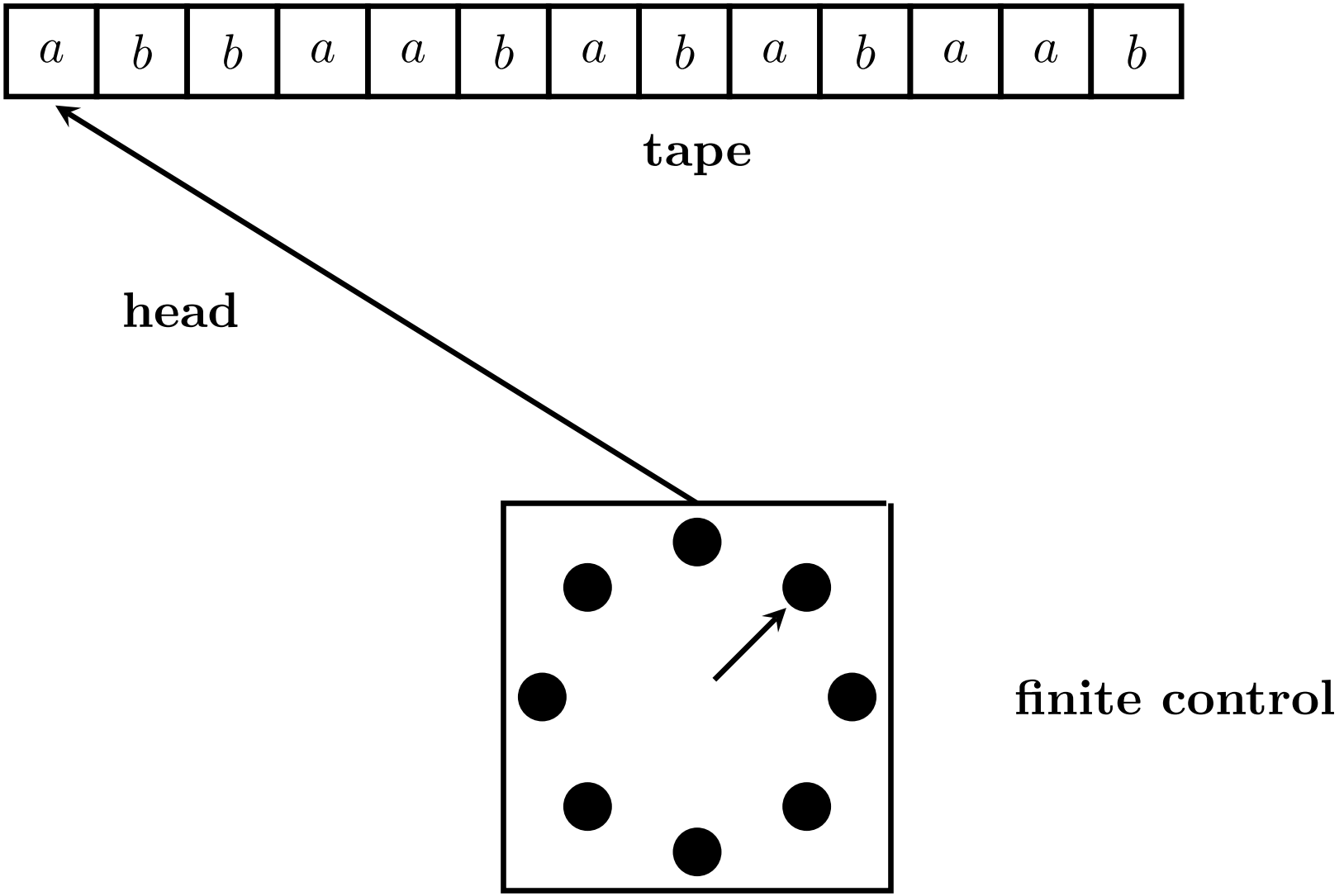
Intuition for DFA

- A **deterministic finite automation (DFA)** is a mechanical machine which reads an input sting one letter at a time and then, after the input is completely read, decides to accept or to reject the input.
- A DFA consists of three parts:
  - a **tape** which used to store the input data;
  - a **tape head** which scans one cell of the tape the information to the finite control, and then moves to the next cell to the right;
  - a **finite control** has a finite number of states. At the beginning of a move, the control is in one of the states. Then it determines, from the current state and the symbol read by the tape head, how the state is changed to to a new state.


a b b a a b a b a b a a b
tape
head
finite control

Change of states

The change of state at each move is governed by a **state transition function** $\delta : Q \times \Sigma \to Q$. At each move, if the finite control is currently at state $q \in Q$ and the symbol read by the tape head is $a \in \Sigma$, then the finite control changes to the new state $q = \delta(q, a)$.

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We also specify two special kinds of states:

- an **initial state** – it is the state which the DFA begins to work,
- a set of **finite states** at which the DFA certifies that th input is in the language.

Formal definition

DFA is a quintuple $M = (Q, \Sigma, \delta, q_0, F)$, where

- Q is the set of states in the finite control;
- Σ is an alphabet for strings in the tape;
- δ is well defined function from $Q \times \Sigma$ to Q;
- q_0 is the initial state;
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In the definition the notion well defined mean that

- function is defined for any input;
- for any input pair there is unique output.

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The set of all strings accepted by a DFA M is denoted by $L(M)$. We also say that the language $L(M)$ is **accepted** by M.

DFA as a labeled digraph

The transition diagram of a DFA is an alternative way to represent the DFA. For $M = (Q, \Sigma, \delta, q_0, F)$, the transition diagram of M is a labeled digraph (V, E) satisfying

- $V = Q$;
- E consists of all edges (p, q) with labed $a \in \Sigma$ iff $\delta(p, a) = q$.

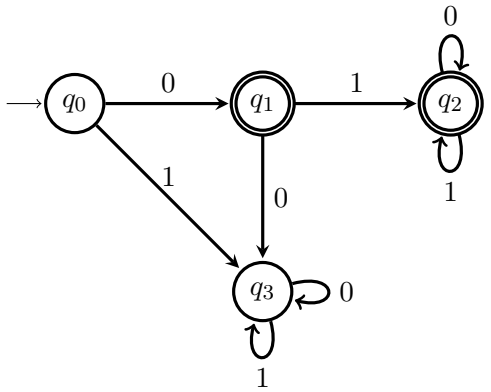
In addition, the initial state of the DFA is pointed to by an arrow withourt a tring vertex, and every final state is denoted by double circles.

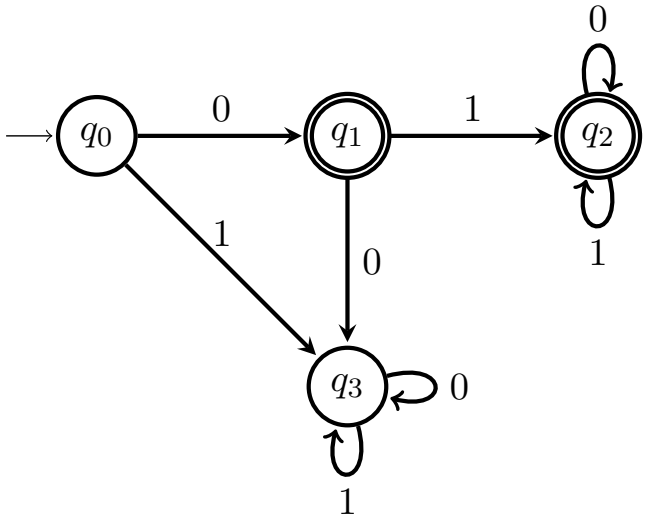
Example

Consider a DFA $M = (Q, \Sigma, \delta, q_0, F)$ where $Q = \{q_0, q_1, q_2, q_3\}$, $\Sigma = \{0, 1\}$, $F = \{q_1, q_2\}$ and δ us the function defined by the following table:

δ	0	1
q_0	q_1	q_3
q_1	q_2	q_3
q_2	q_2	q_2
q_3	q_3	q_3

Then, the transition diagram of M is as the following graph

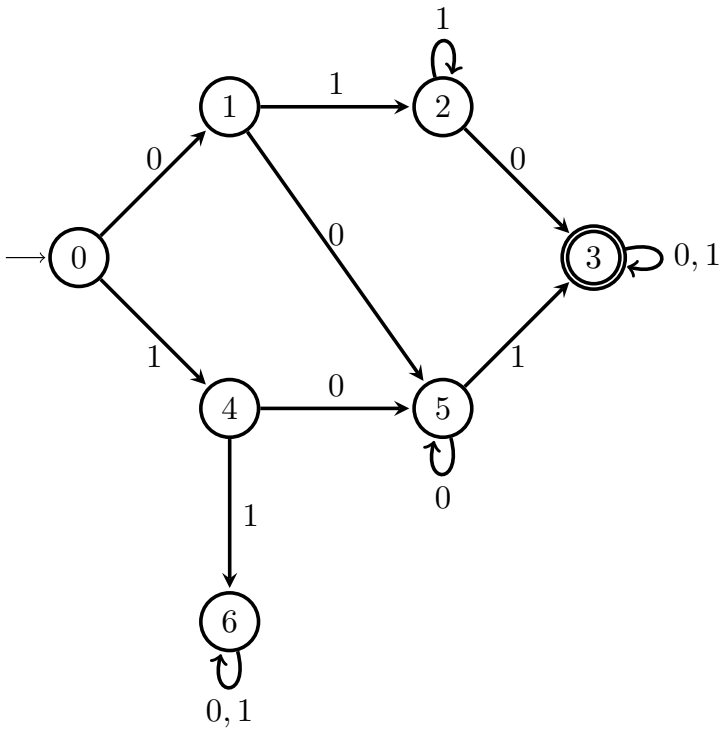




Consider the DFA given as above. Determine weather M accepts the following strings:

- ε;
- 000;
- 111;
- 010;

What is the language $L(M)$ accepted by the DFA.



Example. What is the language $L(M)$ accepted by the DFA M of the figure left (Write its regular expression)?

Theorem

If a language L is accepted by a DFA, then L is regular language.

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If a language L is accepted by a DFA $M_0 = (Q, \Sigma, \delta, q_0, F)$, then the language $\overline{L}$ is accepted by a DFA $M_1 = (Q, \Sigma, \delta, q_0, Q \setminus F)$.

Examples

Construct DFA for which accepted the following languages:

- the set of all binary strings beginning with prefix 101;

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- the set of all binary strings beginning with prefix 101;
- the set of all binary strings having a substring 101;
- the set of all binary strings ending with 101;
- the set of all binary expansions of positive integers which are congruent to zero modulo 5;

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- the set of all binary strings having a substring 101 or ending with 11;

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- the set of all binary strings having a substring 101 and ending with 11;

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- the set of all binary expansions of positive integers which are congruent to zero modulo 5;
- the set of all binary strings having a substring 101 or ending with 11;
- the set of all binary strings having a substring 101 and ending with 11;
- the set of all binary strings in which every block of four consecutive symbols contains a substring 01.

Nondeterministic Finite Automata (NFA)

Definition

A NFA $M = (Q, \Sigma, \delta, q_0, F)$ is defined in the same way as a DFA exept that in NFA multiple-state transformation and ε-transformations are allowed, and definition of accepting are different (we will say about it)

Therefore, the transition function δ is, formally, a function of the form

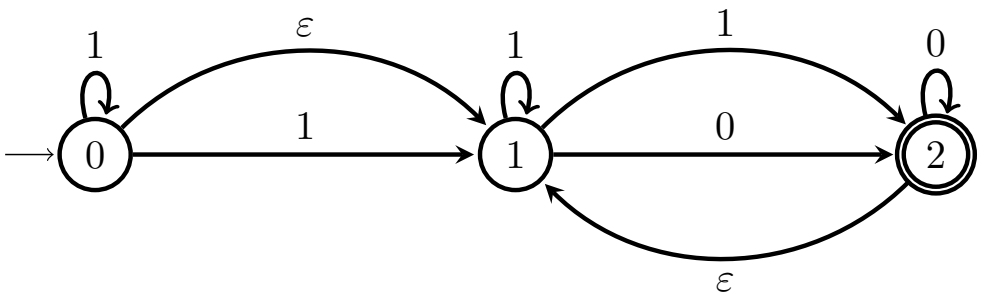
$$\delta : Q \times (\Sigma \cup \varepsilon) \to 2^Q,$$

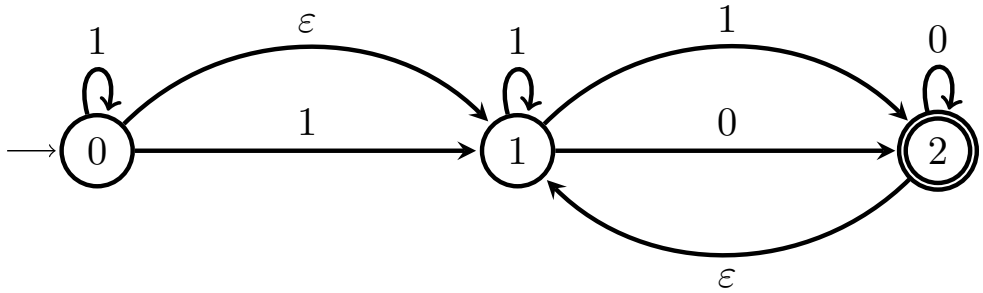
where 2^Q denotes the collection of all subsets of Q.

The following is an example of a transition function of an NFA:

δ	0	1	ε
q_0	$\emptyset$	$\{q_0, q_1\}$	$\{q_1\}$
q_1	$\{q_2\}$	$\{q_1, q_2\}$	$\emptyset$
q_2	$\{q_2\}$	$\emptyset$	$\{q_1\}$

The NFA's, like DFA's, can also be represented by transition diagram.





On an input x, an NFA may have more than one computation path. For example, for NFA M shown in figure above, the input 01 has the following three computation paths:

$$q_0 \xrightarrow{\varepsilon} q_1 \xrightarrow{0} q_1 \xrightarrow{\varepsilon} q_1 \xrightarrow{1} q_1,$$
$$q_0 \xrightarrow{\varepsilon} q_1 \xrightarrow{0} q_2 \xrightarrow{\varepsilon} q_1 \xrightarrow{1} q_2,$$
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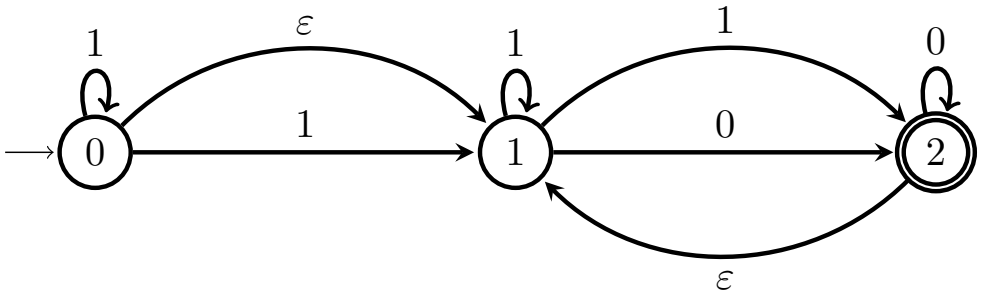

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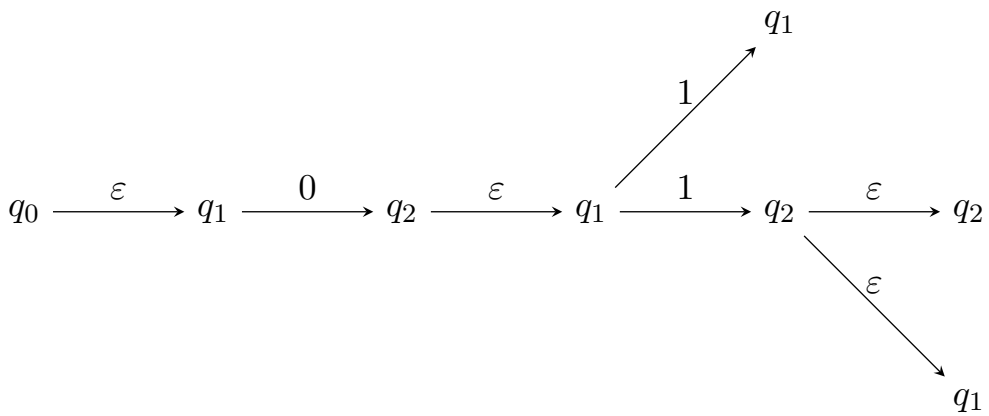

How do we define the notion of an NFA accepting an input? We say that the NFA accepts an input x uf at least one computation path on x leads to a final state. Fir instance, in the last example, since the second computation path ends at state $q_2 \in F$ after the NFA reads the input 01, the string 01 is accepted by the NFA.

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For instance, graph representation for regular expressions is a graph for NFA automata, i.e.

Theorem

If L is regular language, then there is NFA which accepted L.

Examples

Find NFA's which accepted the following languages:

- the set of all binary stings having substring 010;

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- the set of all binary strings beginning with 010 or ending 110;

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- the set of all binary strings with at least two occurrences of substring 01 and ends with 11;

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Find NFA's which accepted the following languages:

- the set of all binary stings having substring 010;
- the set of all binary strings beginning with 010 or ending 110;
- the set of all binary strings with at least two occurrences of substring 01 and ends with 11;
- the set of all binary strings of form 0^n10^m where $n = m(\mod 5)$.